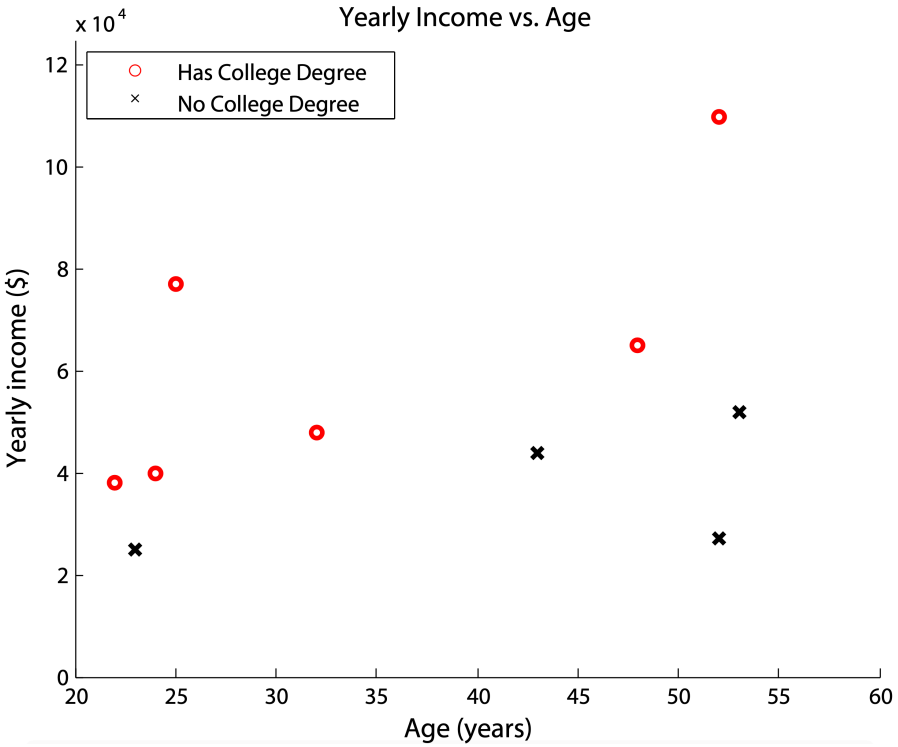
1. Consider the problem of predicting if a person has a college degree based on age and salary. The table and graph below contain training data for 10 individuals.

Age	Salary	College Degree
24	40,000	Yes
53	52,000	No
23	24,000	No
25	77,000	Yes
32	48,000	Yes
52	108,000	Yes
22	38,000	Yes
43	44,000	No
52	27,000	No
48	67,000	Yes



a. (1 points) Build a decision tree for classifying whether a person has a college degree by greedily splitting attributes that maximize information gain. Assume that the threshold for each split is the middle value of the attribute, i.e., ½(min+max), in the current subset. Show the information gain at each split and draw the decision boundary of your learned tree in the figure.

b. (1 points) A multivariate decision tree is a generalization of univariate decision trees, where more than one attribute can be used in the decision rule for each split. That is, splits need not be orthogonal to a feature's axis.

For the same data, provide a multivariate decision tree where each decision rule is a linear classifier that makes decisions based on the sign of $\alpha x_{age} + \beta x_{income} - 1$.

Draw your tree including the α, β and the information gain for each split. Draw the decision boundary.

c. (1 points) Draw the decision boundaries for the 1-nearest neighbor classifier assuming that we are using standard Euclidean distance to compute the nearest neighbors.

2. (2 points) Naive Bayes Classifier is widely used for text classification. In this problem you are given a training set of students' feedback about a canteen with corresponding labels indicating whether the students like or dislike it.

Review	**Sentiment**
The chef must taste dead flies	Dislike
Not delicious but cheap	Dislike
Where can I complaint for food with flies?	Dislike
Not cheap but delicious	Like
Super nice, I like the delicious fish and chicken	Like
The food suits my taste very well	Like

Now given a new review "very delicious and cheap" we would like to classify them into two categories using Naïve Bayes algorithm. Please provide the whole computation process.